TermMap

A clinical term mapping engine for ICD-10-CM and SNOMED CT | HiLabs TermMap Challenge | Repository: github.com/brindoxx/HiLabs-TermMap | Demo video: drive.google.com/file/d/1MOzgHRKtS2zhc2zYM20REIGlclfuRndI

68.9%	53.8%	45.1% / 42.2%	88.9%
ICD-10-CM exact match (held-out)	SNOMED CT exact match (held-out), headline	long tail: gold code unseen in training (ICD / SNOMED)	ICD accuracy when both systems' predictions agree (36% of terms)

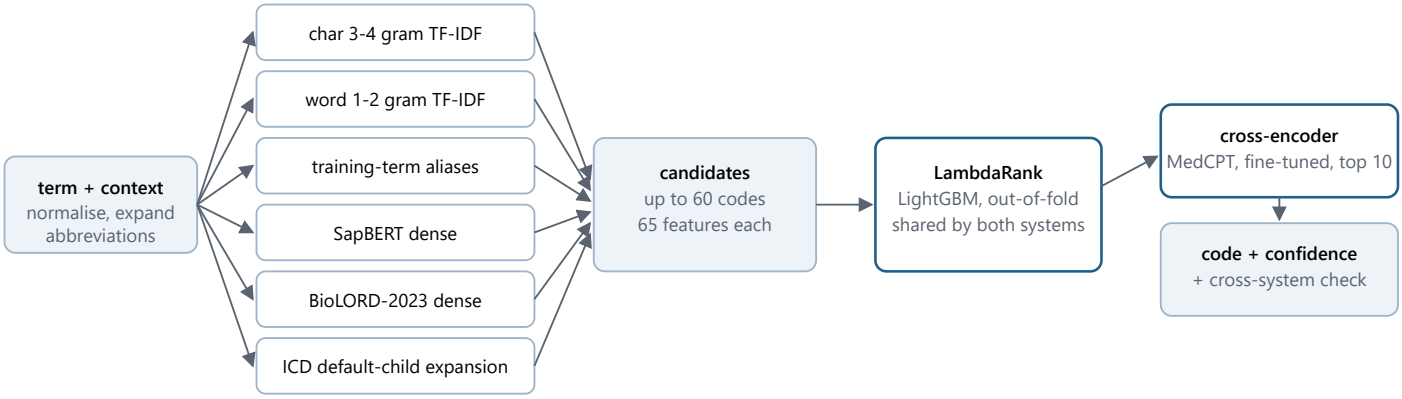
Held-out protocol: 80/20 random split of the 9,646 labelled rows (seed 13); 472 ICD-10-CM and 1,529 SNOMED CT test terms that no retrieval index, prior, reranker or fine-tuned model saw. Blend weights were chosen with cross-fitting (selected on one half of the test terms, scored on the other); the cross-fitted and reported numbers coincide.

1. Approach

We treated the task as entity linking into two very large label spaces (98k ICD-10-CM codes, 382k SNOMED CT concepts), where nearly every input is new: in a random split only about 1% of test terms also appear in training, while 76% of test ICD-10-CM codes and 53% of test SNOMED CT codes do. Memorising terms is therefore useless, but the coding team's habits (which codes they choose, and how they resolve ambiguity) are learnable. That observation shaped the design: a high-recall retrieval stage over the full dictionaries, followed by learned rerankers trained on the coding team's own decisions. Each system is predicted independently with its own retrieval, priors and ranking. Neither is derived from the other.

Scope: all entity categories in the labelled data are handled (diagnoses, but also SNOMED CT procedures, labs, encounters and devices), since the evaluation file's composition is not known. The `subconcept` column is used when present, and a second ranker variant without it is trained so the engine does not depend on it.

2. Architecture



Lexical retrieval and ranking run on CPU. Dense description embeddings and the cross-encoder fine-tune were computed on a T4 GPU (`notebooks/`); at prediction time the engine reads precomputed hits or encodes new queries on CPU. Model locations: encoders and cross-encoder under `artifacts/`, ranker and coding-team crosswalk in `artifacts/engine.joblib`.

3. Design decisions

- **Hybrid retrieval, six sources.** Character n-grams catch OCR damage and run-together words (*chronickidney disease stag 3*); word n-grams catch rare exact tokens; dense encoders catch synonyms with no lexical overlap. Each source added measurable accuracy (Table 1). Training terms are indexed as aliases of their gold codes so local shorthand is retrievable.
- **Context-settled abbreviations.** Each reading of an ambiguous abbreviation carries cue words (right atrium: *echo, tricuspid, rv*; rheumatoid arthritis: *methotrexate, synovitis*). When the chunk supports exactly one reading, the term is searched and scored under that reading, so a frequent reading cannot outvote the context. On the held-out split this leaves ICD-10-CM at 68.9 and moves the SNOMED CT ranker stage from 49.6 to 49.7.
- **Learning to rank, not thresholds.** Lexical and clinical distance diverge exactly where errors happen (*hypertension* vs *pulmonary hypertension*). The ranker sees, per candidate: scores, gaps and ranks from every source; fuzzy similarity; token coverage in both directions; **unexplained description tokens and the share explained by the chunk context**; qualifier agreement (laterality, acuity, history-of, negation, type, stage, encounter); coding-team priors for the code and its ICD block or SNOMED semantic tag, also conditioned on subconcept; ICD leaf, depth and unspecified flags. LambdaRank optimises the order within each term's candidate group.
- **Specificity policy.** Specificity is an explicit, learned choice: a retrieved non-billable ICD category contributes its *unspecified* or *without complication* children, and the unexplained-token features penalise detail the text does not support unless the context supplies it. 81.1% of ICD-10-CM predictions land in the correct 3-character block; a third of remaining ICD errors are right-category, wrong-specificity.

- **Leakage control.** Aliases and priors come from training data, so the ranker is trained on out-of-fold candidates (5 folds), and the cross-encoder is fine-tuned only on those out-of-fold candidate groups. Live cross-encoder scoring is disabled during evaluation.
- **Fine-tuned cross-encoder as the second stage.** MedCPT-Cross-Encoder, fine-tuned on 7,643 groups (gold description plus 15 hard negatives from our own candidate lists, listwise loss), re-scores the top 10 with the term, subconcept and context window. It is weaker than the ranker alone (59.1% / 48.7%) but the two make different mistakes; the log-linear blend gains 3.9 and 4.2 points.

Table 1. Stage-by-stage held-out accuracy	ICD-10-CM	SNOMED CT
Lexical retrieval + aliases, gradient boosting per system	54.0	41.7
+ shared ranker, subconcept priors, LambdaRank objective	54.5	43.0
+ SapBERT dense retrieval	61.7	49.1
+ BioLORD-2023 dense retrieval	65.0	49.6
+ fine-tuned MedCPT cross-encoder (submitted configuration)	**68.9**	**53.8**
Zero-shot Qwen2.5-7B adjudication instead, measured at the SapBERT stage (cross-fitted)	58.7	53.4

Table 2. Diagnostics (submitted configuration)	ICD	SNOMED
Gold in candidate set / in top 10	90.3 / 87.1	86.7 / 78.6
Accuracy, gold code seen in training	76.3	64.1
Accuracy, gold code unseen (long tail)	45.1	42.2
Share of test terms with unseen gold code	23.9	46.8
ICD 3-character block accuracy	81.1	
Without the subconcept column (ranker stage)	64.8	49.1

Table 3. Cross-system agreement as confidence	share	ICD acc.
Predicted pair used by the coding team	36.2	88.9
Codes paired by the team, but differently	9.3	31.8
No pairing evidence	54.5	61.9

For SNOMED CT targets: 73.5% accuracy on agreeing pairs (12.1% of terms) against 51.8% when they disagree.

4. Error analysis

Every held-out error was assigned to the first matching failure mode (`scripts/error_analysis.py`; the abbreviation, noise and multi-concept categories are heuristic).

Failure mode	ICD errors	SNOMED errors	Typical example (term: gold vs predicted)
Gold never retrieved	46 (31%)	203 (29%)	*activity as tolerated*: Ambulation without limitation vs Activity tolerance
Right ICD block, wrong specificity	47 (32%)		*acute on chronic respiratory failure*: J96.20 vs J96.00; *adenopathy*: R59.9 vs R59.0
Right words, wrong SNOMED hierarchy		143 (20%)	*accu checks*: Capillary blood glucose measurement (procedure) vs Blood glucose monitoring (regime/therapy)
Abbreviation / shorthand	11 (7%)	46 (7%)	*dm1*: E10.9 vs E11.9; *100% af*: Persistent AF vs Atrial fibrillation
OCR / typing noise	15 (10%)	38 (5%)	*a r l5 and s1 tfesi*: lumbar transforaminal injection vs spine-level parent
Several ideas in one term	6 (4%)	38 (5%)	*impaired sensation in feet 2/2 neuropathy*: R20.2 vs E11.40
Local convention far from wording	11 (7%)	93 (13%)	*bacterial infection*: B99.9 vs A49.9; *delayed gastric emptying*: K30 vs K31.84
Other close near-miss	11 (7%)	145 (21%)	*acute ischemia*: Acute myocardial ischemia vs Ischemia

- **Retrieval is the ceiling for a third of errors.** These are mostly paraphrases of functional status, therapy and planning concepts in SNOMED CT, where neither spelling nor a general biomedical encoder reaches the coding team's chosen concept.
- **Specificity errors are symmetric.** We see both over-coding (R59.0 localized for *adenopathy*) and under-coding (I50.9 for *advanced heart failure*, gold I50.84 end stage), so the policy is not biased one way. The fix needs better qualifier reading, not a global bias shift.
- **SNOMED hierarchy choice is convention.** The same words exist as a finding, an observable and a procedure; which one the team picks depends on the entity category and on habit. Subconcept priors and the cross-encoder each recover part of this.
- **Label and version drift.** 5.3% of ICD-10-CM gold descriptions in the training file (123 rows) are not descriptions in the supplied dictionary (e.g. *Nonrheumatic aortic valve insufficiency* for I35.1), pointing to a different release behind part of the labels.
- **Practical ceiling.** Some gold labels cannot be derived from the term and its context alone (*bacterial endocarditis* coded A49.01). Inter-annotator agreement on the evaluation set would quantify how much of the remaining gap is reachable.

5. Innovations

- **Coding-team crosswalk as a confidence signal.** 2,124 labelled terms carry both an ICD-10-CM and a SNOMED CT gold code. From them we learn the team's own pairings and check whether the two independent predictions form a known pair. Agreeing

pairs (36% of terms) are 88.9% accurate for ICD-10-CM; disagreeing pairs 31.8%. Used as a soft prior it did not hold up under cross-fitting, so it is used for confidence and review routing, not to change predictions.

- **One ranker for two terminologies.** A single LambdaRank model with a system indicator is shared by both systems, so the smaller ICD-10-CM set benefits from the SNOMED CT examples.
- **Hard negatives from our own retriever.** The cross-encoder is fine-tuned on exactly the confusions our pipeline makes, built out-of-fold so no training group contains its own term as an alias.
- **Calibrated output.** Confidence plus the cross-system check support abstention: abstaining on the 5% of terms with the lowest ranker confidence raises accuracy by 3.0 points (ICD-10-CM) and 2.1 points (SNOMED CT).

6. Tried and abandoned; challenges and learnings

- **Zero-shot LLM adjudication** (Qwen2.5-7B-Instruct, 4-bit, letter-probability scoring with retrieved in-context examples): +4.4 points on SNOMED CT but 2 to 3 points worse on ICD-10-CM, where it overrides the team's conventions with textbook choices. The fine-tuned cross-encoder helps both systems at a fraction of the cost, so it replaced the LLM.
- **LoRA fine-tuning of an LLM selector** (Qwen2.5-3B, QLoRA on out-of-fold prompts) is implemented in `notebooks/lora_selector.ipynb`; GPU quota ran out before it could be evaluated.
- **Gold display texts as extra synonyms** lowered ICD-10-CM accuracy by one point and were removed.
- **Swapping the reranker learner** (histogram GBM, LightGBM binary, LambdaRank) changed accuracy by about one point; every large gain came from a new signal (dense retrieval, the cross-encoder), not from the learner.
- **Constraints.** The data cannot leave our environment, so every model is open-weight and self-hosted; no hosted LLM API was used.

7. Next steps for the remaining gap

- Raise candidate recall, the hard ceiling: fine-tune SapBERT contrastively on the team's term-to-code pairs (one model per fold), and add an LLM rewrite of noisy or bundled terms before retrieval.
- Apply LLM adjudication only to low-confidence or cross-system-disagreeing terms, which bounds cost and targets the 32%-accuracy group.
- Use the official SNOMED CT to ICD-10-CM map and the SNOMED IS-A hierarchy (UMLS licence) for hierarchical features and partial credit.
- Abbreviations with numerals and qualifiers (*dm1*, *100% af*) need a context-aware expansion model rather than a lookup table.

8. Reproducibility, cost and external resources

`python -m termmap predict --input <evaluation file> --output outputs/predictions.csv` reproduces the predictions end to end. Every split and model uses seed 13, LightGBM runs in deterministic mode, dense search and cross-encoder scoring are deterministic, and no sampling is used anywhere.

Latency per term, both systems, unseen terms (4-core laptop CPU, no GPU)	median	p95
50 random labelled terms	3.4 s	4.6 s
8 hard cases	2.7 s	3.1 s
All 58 terms (one-time model load: 30 s)	**3.2 s**	**4.6 s**

Stage, per term	approx. time
Lexical retrieval, features, ranker	30 ms
Dense encoding and search (2 encoders x 2 systems)	0.3 to 0.5 s
Cross-encoder (10 candidates x 2 systems)	2 to 3 s

Latency fell from about 90 s to about 3 s per term by holding the description embeddings in RAM instead of memory-mapping them (which re-read 1 GB from disk per term on a memory-constrained machine), merging top candidates chunk by chunk, and loading every model at start-up; predictions are unchanged. Optional int8 embeddings halve memory (2.4 to 1.2 GB) with 97 to 99% identical top-1 dense hits; int8 encoders and cross-encoder were rejected because they changed 16 to 74% of top choices. At production scale: GPU batching, an approximate nearest-neighbour index (FAISS IVF-PQ or HNSW), running the cross-encoder only on low-confidence terms, a term-and-context cache, and a distilled cross-encoder served with ONNX Runtime.

Resource	Use	Licence
ICD-10-CM and SNOMED CT US descriptions (supplied files; SNOMED export dated 24 Sep 2026)	answer space, retrieval	CDC/NCHS public domain; SNOMED CT affiliate licence
`cambridgeltl/SapBERT-from-PubMedBERT-fulltext`	dense retrieval	Apache 2.0
`FremyCompany/BioLORD-2023`	dense retrieval	see model card
`ncbi/MedCPT-Cross-Encoder`	cross-encoder, fine-tuned	see model card
`Qwen/Qwen2.5-7B-Instruct`, `Qwen/Qwen2.5-3B-Instruct`	LLM experiments	Apache 2.0 / Qwen licence
LightGBM, scikit-learn, RapidFuzz, pandas, PyTorch, Transformers, PEFT	modelling	MIT / BSD / Apache 2.0

No official crosswalk or UMLS release was used. The coding-team crosswalk is derived from the supplied labelled data only.